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Sequential application of small molecule therapy enhances chondrogenesis and angiogenesis in murine segmental defect bone repair

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Abstract

The increasing incidence of physiologic/pathologic conditions that impair the otherwise routine healing of endochondral bone fractures and the occurrence of severe bone injuries necessitate novel approaches to enhance clinically challenging bone fracture repair. To promote the healing of nonunion fractures, we tested an approach that used two small molecules to sequentially enhance cartilage development and conversion to the bone in the callus of a murine femoral segmental defect nonunion model of bone injury. Systemic injections of smoothened agonist 21k (SAG21k) were used to stimulate chondrogenesis through the activation of the sonic hedgehog (SHH) pathway early in bone repair, while injections of the prolyl hydroxylase domain (PHD)2 inhibitor, IOX2, were used to stimulate hypoxia signaling-mediated endochondral bone formation. The expression of SHH pathway genes and *Phd2* target genes was increased in chondrocyte cell lines in response to SAG21k and IOX2 treatment, respectively. The segmental defect responded to sequential systemic administration of these small molecules with increased chondrocyte expression of PTCH1, GLI1, and SOX9 in response to SAG and increased expression of hypoxia-induced factor-1 α and vascular endothelial growth factor-A in the defect tissues in response to IOX2. At 6 weeks postsurgery, the combined SAG–IOX2 therapy produced increased bone formation in the defect with the bony union over the injury. Clinical significance: This therapeutic approach was successful in promoting cartilage and bone formation within a critical-size segmental defect and established the utility of a sequential small molecule therapy for the enhancement of fracture callus development in clinically challenging bone injuries.

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AUTHOR CONTRIBUTIONS

Charles H. Rundle performed animal procedures, acquisition, analysis of in vivo results, and manuscript preparation. Gustavo A. Gomez performed the acquisition and analysis of immunohistochemistry data. Sheila Pourteymoor performed the acquisition and analysis of in vitro and molecular expression data and performed the animal procedures. Subburaman Mohan contributed to research design, analysis and interpretation of data, and manuscript preparation. All authors have read and approved the final submitted manuscript.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Keywords

angiogenesis; chondrogenesis; PHD2 antagonist; segmental defect; sonic hedgehog agonist

1 | INTRODUCTION

Bone injuries result from a wide variety of causes and impaired bone healing is a serious health and economic problem. Of the estimated 7.9 million fractures that occur annually in the US, approximately 10% of fractures exhibit delayed healing, with 100,000 progressing to delayed or nonunion healing and resulting in serious pain and disability.¹ There is, therefore, a compelling need to develop effective therapies to avoid nonunion bone healing in severe bone injuries or in physiological conditions that complicate healing, such as osteoporosis, cancer, diabetes, or simply age.²

Normal healing of long bone injuries in adults involves the formation of a cartilage template on which new bone is formed,³ recapitulating endochondral bone formation of embryonic development to produce bony union at the fracture site.⁴ This process is coordinated by three main classes of growth factors: proinflammatory cytokines and growth factors, pro-osteogenic factors, and angiogenic factors.^{3,5-7} Many investigations have characterized the expression and function of growth factors and signaling molecules in normal fracture repair, with the objective of augmenting their functions through therapeutic intervention to improve impaired bone healing.^{8,9}

The current therapies for nonunion fracture healing use bone grafting, biosynthetic polymers, mesenchymal stromal cells, and osteogenic growth factors to promote bone formation at the fracture site. However, their impact on bone healing remains uncertain.¹⁰ Additionally, studies have noted concerns about ectopic bone formation with the bone morphogenetic proteins.¹¹ There remains an urgent need to develop a safe and effective but low-cost therapy that promotes the healing of nonunion skeletal defects.¹²

The primary sources of osteoblasts regulating endochondral ossification during skeletal development and fracture repair are mesenchymal cells originating from the periosteum and blood^{4,13,14} that proliferate and differentiate to the osteogenic lineage. Fracture callus osteoblasts have been traditionally thought to be imported to replace hypertrophic chondrocytes, but there is considerable evidence supporting chondrocyte transdifferentiation to osteoblasts. We and others have demonstrated that chondrocytes undergo transdifferentiation into bone-matrix-producing osteoblasts and contribute to endochondral ossification during normal skeletal development and fracture repair.^{13,15-20} Because differentiating chondrocytes can provide an important source of osteoblasts during fracture repair, we propose transdifferentiation as a therapy to enhance bone formation in a clinically challenging model of bone repair.

The thyroid hormone is a key physiological regulator of endochondral bone formation during normal skeletal development, since short stature, impaired bone maturation, and delayed bone healing are well-established features of hypothyroidism.²¹ Furthermore, thyroid hormone is indispensable for the initiation and progression of secondary ossification

at the long bone epiphyses.²² We have found that triiodothyronine (T3) promotes an expansion of immature chondrocytes expressing type-2 collagen (*Col2*) and aggrecan (*Acan*) by stimulating sonic hedgehog (SHH) signaling.¹⁷ The activation of SHH signaling to differentiate mesenchymal cells into chondrocytes that form a cartilaginous template for ossification is critical for endochondral bone formation in development.^{19,23–25}

Bone healing studies have established that SHH is expressed at the fracture site in both endochondral and intramembranous bone formation.^{24,26–28} Functionally, SHH agonists improve intramembranous and endochondral bone repair.^{27,29,30} Interestingly, SHH might be important for the repair of large-scale rib defects³¹ and could produce sufficient cartilage for femoral segmental defect repair.

We have previously demonstrated that hypoxia signaling pathways promote chondrocyte hypertrophy and transdifferentiation to osteoblasts.¹⁷ Hypoxia-induced factor (HIF)-1 α is a critical mediator of angiogenesis, as deficient HIF-1 α signaling produces a complete failure of chondrocyte hypertrophy and severe defects in endochondral bone formation.³² HIF-1 α is inhibited by prolyl hydroxylase domain (*Phd*)-2 functions. Conditional deletion of *Phd2* gene expression in chondrocytes produces a large increase in trabecular bone mass caused by the HIF-1 α -mediated promotion of chondrocyte differentiation and conversion into bone-forming osteoblasts.³³ Conversely, exogenous PHD inhibitors promote bone formation and callus vascularity in fracture repair.³⁴ Thus, there is compelling evidence that angiogenic signaling pathways could be modulated by PHD2 functions and that they could promote bone formation through the transdifferentiation of cartilage chondrocytes for defect healing.^{13,15}

This study describes a therapeutic strategy for segmental defect healing that uses the sequential application of small molecule compounds to first promote SHH expansion of chondrocytes to a soft callus cartilaginous template, and subsequently promote angiogenesis and cartilage hypertrophy for chondrocyte-to-osteoblast transdifferentiation and bony callus formation. These objectives are achieved by sequential applications of (1) SAG21k (“Smoothened Agonist”) 21k, a potent small molecule agonist derivative of SAG that binds to the SHH receptor Smoothened (SMO) and activates the SHH pathway, enhancing cartilage formation,³⁵ and (2) IOX2, a small molecule antagonist of PHD2 that promotes angiogenesis and chondrocyte replacement by osteoblasts through HIF-1 α signaling.³⁶

2 | METHODS

2.1 | Cell culture

In vitro gene expression in response to SAG21k or IOX2 functions was determined in the chondrocyte cell line ATDC5 (American Type Culture Collection) or in epiphyseal chondrocytes isolated from 5-day-old C57BL/6 mice by digestion with collagenases I and II.³⁷ ATDC5 and primary chondrocytes were maintained in F12/Dulbecco’s modified Eagle’s medium (DMEM) supplemented with 5% or 10% fetal bovine serum, respectively. Gene expression was measured by real-time reverse transcription polymerase chain reaction (RT-PCR) using gene-specific primers (Supporting Information: Table 1).

To compare SAG and SAG21k enhancement of chondrocyte proliferation, each was titrated for enhancement of expression of gene markers of proliferation. Chondrocyte cultures were serum-starved for 24 h in F12/DMEM (0.1% bovine serum albumin) to synchronize the cell cycle, titrated with SAG or SAG21k to a maximum of 100 nM for 72 h and gene expression then analyzed.

A SAG21k dosage of 1 nM provided maximum expression of the *Ki67* and *Ccnd1* proliferation markers. To determine SAG21k functions in chondrocyte development, this dosage was applied to primary chondrocytes in culture for 72 h. The expression of the *Shh* pathway receptor “patched” gene (*Ptch1*), transcription factor target genes (*Gli1*, *Gli2*), and genes functioning in chondrocyte development (*Sox9*, *Col2*, *Col10*), remodeling (matrix metalloproteinase 13, *Mmp13*), and mineralization (alkaline phosphatase, *Alp*) were then determined.

To demonstrate that IOX2 promotes *Hif-1 α* functions to enhance chondrocyte hypertrophy and angiogenesis, cultures of ATDC5 and primary epiphyseal chondrocytes were evaluated in response to the PHD2 inhibitor for the expression of marker genes of chondrocyte development (*Col2*, *Col10*, and *Acan*) as well as glycolysis (glucose transporter, *Glut1*) and angiogenesis (vascular endothelial growth factor-A, *Vegfa* and erythropoietin, *Epo*). Cultures were synchronized at 70% confluence and then incubated with 20 μ M IOX2 or dimethylsulfoxide (DMSO) vehicle control for 72 h before gene expression analysis. Additionally, chondrocytes isolated from a *Phd2*-loxP mouse line in which the *Phd2* gene was deleted through adenoviral-improved Cre (Ad-iCre) application in vitro were analyzed for the expression of these genes and for the osteogenic gene osterix (*Osx*). Cultures were maintained in α minimum essential medium (10% fetal bovine serum) and transduced at 20% confluence with the Ad-iCre deleter or Ad-green fluorescent protein at a multiplicity of infection of 5. Following a 24-h incubation, the media was changed for the 72 h IOX2 incubation and subsequent analysis of gene expression.

2.2 | RNA isolation and quantitative real-time RT-PCR

Total RNA (EZNA Total RNA Kit; VWR) was isolated from the cultures using TRI Reagent (Molecular Research Center Inc.) and reverse-transcribed to complementary DNA using the Superscript IV Kit (Invitrogen). Real-time RT-PCR was performed using the Applied Biosystems SYBR Green Mastermix. PCR cycling conditions used an annealing temperature of 60°C for 40 1-min cycles. Target gene expression was normalized to the peptidyl-prolyl isomerase A housekeeping gene and changes in expression compared to controls by the $\Delta\Delta C_t$ approach. The primer sequences used to amplify the target genes are presented in Supporting Information: Table 1.

2.3 | Animals

All procedures involving animals were conducted in accordance with the Animal Welfare Act and approved by the local Institutional Animal Care and Use Committee. Ten-week-old specific pathogen-free C57BL/6J female mice were used in all procedures. Surgical procedures were performed in a dedicated surgical facility.

2.4 | Surgical procedures

Anesthesia was administered by the inhalation of 3% isoflurane in 0.7 L/min oxygen using a nonrebreathing circuit. An adequate plane of anesthesia was monitored through the breathing rate and paw reflex.

A 1.8 mm segmental defect was surgically produced in the femoral midshaft. This method adapts the intramedullary stabilization from the closed femur model to stabilize a larger bone defect than the closed fracture but exhibits severely impaired healing that progresses to nonunion.³⁸ Bony union in a closed fracture of the murine femur normally occurs between 3 and 4 weeks, with fracture callus remodeling then completing healing. Critical-size endochondral bone defects of more than 2.0 mm result in nonunion healing.³⁹ The 1.8 mm distance used in this study produced a highly reproducible defect with severely delayed healing that could be used to evaluate the therapeutic enhancement of bone repair. Analgesia consisted of 60 µg/kg buprenorphine delivered subcutaneously, beginning para-operatively and repeated every 12 h for 2 days, and longer if discomfort persisted, as determined by activity, grooming, and weight-bearing.

All animals were monitored weekly in vivo X-ray examination during the 6-week healing period. A total of 42 animals underwent the segmental defect procedure. Following surgery, seven individuals were excluded from the defect healing analysis when defect compression became evident, but their therapy applications were completed and they contributed to the contralateral femur analysis.

2.5 | Application of therapy

Following each surgery, defects were confirmed by X-ray examination, and the mice were randomly divided into three treatment groups and one control group according to defect quality. Treatment groups received SAG21k, IOX2, or a combined SAG21k:IOX2 therapy, and the control mouse group received the DMSO vehicle. All applications were by intraperitoneal injection. Each treatment group totaled nine animals and the control group eight animals, group sizes based upon previous microcomputed tomography (microCT) analysis of bone formation in closed femur fracture repair. To minimize confounders, the individuals contributing to each treatment group were generated from different surgeries.

The therapeutic window for delivery of SAG21k and IOX2 is based on the sequence of fracture callus tissues that develop during endochondral bone repair and our current understanding of hedgehog and hypoxia signaling in their regulation.³ Briefly, we choose to administer SAG21k for 7 days beginning 1-day postsurgery, based upon observations that the expression of *Shh* and its downstream target, *Gli1*, were detected by in situ hybridization and RT-PCR in proliferating cells of the fracture callus at 2 days postfracture, but had attenuated by 12 days.²⁴ SAG has been found to be effective in activating SHH signaling in mice at a systemic dosage of 5–20 mg/kg/day.^{40,41} However, since we found SAG21k to be several fold more active than SAG, we chose an intraperitoneal dosage of 5 mg/kg/day SAG21k in 50 µl.

The IOX2 dosage was determined from preliminary observations of IOX2 promotion of trabecular bone formation and its enhancement of hypoxia signaling in other

investigations.⁴² To induce chondrocyte hypertrophy and osteoblast development, we administered IOX2 at a dosage of 20 mg/kg/day in 50 μ l for 3 weeks starting at 8 days postsurgery. Vehicle control injections of 50 μ l were applied for 3 weeks after the first week of SAG21k injections in the SAG21k group, or in the first week before the 3 weeks of IOX2 applications in the IOX2 group. The vehicle was applied to the control animal group for the entire 4 weeks postsurgery. The combination therapy corresponded with the individual therapies and consisted of SAG21k applied for the first 7 days and then IOX2 applied from 8 days through 3 weeks postsurgery. Tissues were harvested for analysis 6 weeks after surgery. The mice were euthanized by carbon dioxide inhalation at 40% volume displacement per minute for 5 min to induce narcosis, and 3 min immersion to ensure apnea.

2.6 | MicroCT analysis of the contralateral femur

The unfractured femurs were examined for bone formation by microCT analysis using the Scanco *VivaCT40* (Scanco Medical AG). The analysis was normalized to the length of the femur. The metaphyseal trabecular bone was “contoured” to exclude the cortical bone and analysis of trabecular bone parameters was performed over a broad range of bone densities (220–1000 mg/cc). A separate analysis of the midshaft cortical bone was conducted at bone densities of 260–1000 mg/cc.

2.7 | Radiologic analysis of healing

Segmental defect healing was monitored by in vivo X-ray examination (Faxitron LX-60; Faxitron) at weekly intervals postsurgery. Defect healing was analyzed at 6 weeks postsurgery for bone formation within the segmental defect by microCT and histology, and for target protein response in the callus by immunohistochemistry. MicroCT analysis determined the bone volume (BV) produced by each therapeutic approach as compared to the vehicle control. The entire defect was scanned and a 2 mm section at the center of the defect was contoured and analyzed by a blinded observer using bone density analysis thresholds adjusted to include lower and higher density callus and cortical bone (250–1000 mg/cc).

2.8 | Histologic analysis of healing

The fracture histology was compared among all individuals in the three treatment groups and the control group of animals. Following microCT analysis, the femurs were decalcified in ethylenediaminetetraacetic acid and paraffin-embedded. Hematoxylin-eosin staining assessed neutrophil infiltrations that would indicate a confounding infection. Cartilage was visualized through a Safranin-Orange stain. Histomorphometry was performed using ImagePro software (Media Cybernetics).

2.9 | Immunofluorescence

Immunofluorescence was used to confirm the expression of pathway target proteins in response to SAG21k function at 5 days postsurgery, or in response to IOX2 inhibition of PHD2 at 4 weeks postsurgery, times when the tissues were expected to respond to each therapy with the expression of its target proteins.³⁷ These procedures were performed by a blinded operator. Two pairs of SAG21k and control individuals and one pair of IOX2 and

control individuals were compared at these times. Primary rabbit polyclonal antibodies to these pathway proteins were applied to the paraffin tissue sections and specific binding was detected by fluorescent secondary anti-rabbit antibodies. Negative control sections omitted the primary antibody. The primary antibodies and test conditions for the SAG21k and the IOX2 pathway proteins are presented in Supporting Information: Table 2, as are antibodies to additional target proteins.

2.10 | Statistical analyses

Data are presented as mean $\pm$ standard error of the mean. Statistical analysis was performed by Student's *T* test and the therapeutic effect was determined to be significant at $p < 0.05$.

3 | RESULTS

3.1 | SAG21k enhances SHH pathway gene expression in chondrocytes

An in vitro titration of the dose-response produced by the SMO agonist SAG in chondrocytes determined that the SAG21k derivative is much more potent than SAG in promoting the expression of the cell cycle gene cyclin D1 (*Ccnd1*, Supporting Information: Figure 1A) and the SHH mediator (*Gli1*, Supporting Information: Figure 1B). For subsequent in vitro gene expression analysis, a dosage of 1 nM SAG21k was used to induce the expression of the *Shh* pathway and cartilage formation genes.

In vitro gene expression measurements confirmed that this SAG21k dosage promoted twofold to threefold increases in the expression of the *Ki67* and *Ccnd1* genes, associated with proliferation, (Figure 1A), as well as increases in the expression of genes that promote cartilage maturation (*Col10*) and mineralization (*Alp*) during callus development, but not callus remodeling (*Mmp13*, Figure 1B). Furthermore, this gene expression coincided with significant increases in the expression of genes of the SHH regulatory pathway; the transcription factor *Gli1*, and *Ptch1*, the cell-membrane-associated inhibitor of SMO-mediated SHH signaling, were both significantly increased. *Gli2* and *Sox9* expression was slightly increased or reduced, respectively (Figure 1C).

3.2 | IOX2 enhances angiogenic and osteogenic gene expression in chondrocytes

The response of the ATDC5 chondrocyte cell line to IOX2 treatment demonstrated that the expression of markers of glycolysis (*Glut1*) and chondrocyte differentiation (*Col10*, *Acan*, and *Vegfa*) was increased, consistent with *Phd2* inhibition and the promotion of *HIF-1 α* -related functions in fracture cartilage differentiation. *Acan* and *Glut1* expression increased approximately twofold (Figure 2A). The expression of genes that promote angiogenesis (*Vegfa* and *Epo*) necessary for osteogenesis was also increased more than twofold in primary chondrocytes by IOX2 (Figure 2B); *Phd2* expression was increased approximately fivefold by IOX2 inhibition of PHD2, consistent with a feedback regulation of expression. Gene expression in response to IOX2 treatments was confirmed following in vitro Ad-iCre-mediated *Phd2* deletion in primary chondrocytes isolated from a *Phd2-loxP* mouse line. This approach reduced *Phd2* expression by 63% in the Ad-Cre transduced cells. *Glut1*, *Vegfa*, and *Epo* expression was upregulated at least twofold relative to control cultures, as was *Osx*

(Figure 2C). These results suggest that the angiogenic and osteogenic gene expression from IOX2 inhibition of PHD2 expression was analogous to *Phd2* deletion.

3.3 | Systemic therapy augmented trabecular bone formation in the unfractured femurs

Because the application of the therapy was systemic, we evaluated bone formation in the unfractured contralateral femur. Examination of the microCT reconstructions (Supporting Information: Figure 2A–D) suggests that IOX2 (Supporting Information: Figure 2C) and the combined SAG21k:IOX2 (Supporting Information: Figure 2D) therapies increased the metaphyseal trabecular bone. MicroCT analysis of the unfractured bones revealed that the SAG21k therapy produced a slight increase in the total tissue volume (TV, $p < 0.05$, Supporting Information: Figure 3A). However, the combined therapy did produce significant changes in the BV ($p < 0.05$, Supporting Information: Figure 3B) and partial BV (BV/TV) compared with the other treatment groups ($p < 0.01$ vs. SAG21k, Supporting Information: Figure 3C). These changes were evident in the trabecular parameters, specifically the connectivity density ($p < 0.05$ vs. SAG21k, Supporting Information: Figure 3D). The combined therapy did not affect the trabecular thickness (Supporting Information: 3E), but the trabecular spacing was reduced ($p < 0.05$ vs. SAG21k, Supporting Information: 3F) through an increase in the trabecular number ($p < 0.05$ vs. IOX2, Supporting Information: 3G). The combined therapy also reduced the ratio of the structure model index compared to SAG21k ($p < 0.005$) and IOX2 ($p < 0.05$), characteristic of a more plate-like trabecular structure (Supporting Information: Figure 3H). None of these parameters in the IOX2 therapy individuals were significantly different from the control and SAG21k therapy. None of the therapies produced a significant effect on the cortical bone, suggesting that their actions were not observed in the limited numbers of osteoblasts present in an unfractured periosteum.

3.4 | Small molecule therapy enhanced bone formation in segmental defect healing

An examination of the three-dimensional microCT reconstruction of the healing defects showed apparent bone formation within the defect gap in response to the different therapies relative to the controls (Figure 3A). Bone formation was evident within the defect gap with the SAG21k therapy (Figure 3B) but not the IOX2 therapy (Figure 3C). The combined therapy, however, produced a robust callus with an apparent bony union at the defect site (Figure 3D).

A microCT analysis of these defects revealed significant changes in the callus TV and BV within the defect gap. When the entire fracture callus was analyzed over bone densities from 250 to 1000 mg/cc, the SAG21k and combined therapies increased the callus TV ($p < 0.05$, Figure 4A) and BV ($p < 0.05$, Figure 4B) which together produced no significant changes in the BV/TV (Figure 4C). Because the IOX2 therapy appeared to augment callus bone at the cortical edges of the defect (Figure 3C), the analysis was then segmented to examine only the higher density bone characteristic of the remodeling callus and cortical bone (570–1000 mg/cc). In this case, there was a highly significant increase in the BV/TV with the IOX2 therapy relative to the SAG212k ($p < 0.01$) and combination ($p < 0.0001$) therapies (Supporting Information: Figure 4).

A histological comparison of the fracture callus anatomy at 6 weeks postsurgery confirmed the bony union in response to the combination therapy, but a significant gap remained when the vehicle control was applied (Figure 5A). Different degrees of bone formation appeared in the defect calluses, with the SAG21k defect exhibiting cartilage (Figure 5B) that by histomorphometric measurement was not significantly different from the controls or IOX2 therapy (Figure 5A,C). Unlike the combination therapy, (Figure 5D), the control and individual therapies retained soft tissue within the defect gap. The combination therapy, therefore, appeared to promote bone formation within the defect.

To confirm the activation of the respective SAG21k and IOX2 pathways, defect calluses were examined at 5 days and 4 weeks postsurgery, respectively, for the expression of the targets of SHH and HIF-1 α pathway activation. SAG21k did indeed produce a dramatic increase in the defect cartilage expression of PTCH1, an SHH membrane receptor (Figure 6A,B), a slight increase in GLI expression (Figure 6C,D) and an obvious increase in the expression of SOX9, a transcription factor for chondrocyte commitment (Figure 6E,F). These results suggest that SAG21k promoted cartilage formation through the SHH pathway early in healing. Additionally, the IOX2 therapy did produce observable increases in HIF-1 α (Figure 7A,B) and VEGFA (Figure 7C,D) in the defect tissues at 4 weeks, suggesting that the HIF-1 α pathway stimulated angiogenesis through VEGFA expression.

4 | DISCUSSION

This study examined a novel combination of small molecule activators of hedgehog and hypoxia signaling designed to sequentially enhance thyroid hormone regulation of cartilage formation⁴³ and its conversion to the bone by vitamin C pathway regulation⁴⁴ for segmental defect healing. Thyroid hormone is well-established in the SHH pathway⁴⁵ regulation of endochondral bone formation^{21,43} and repair.⁴⁶ SAG is, therefore, expected to promote the differentiation of mesenchymal stem cells toward the chondrocyte lineage and the expansion of cartilage, which is normally insufficient in the segmental defect.³⁸

We confirmed in vitro that SAG21k⁴⁷ was very potent in stimulating *CcnD1* expression in chondrocytes (Supporting Information: Figure 1); it increased the expression of markers of chondrocyte development but reduced the expression of mature chondrocyte markers, consistent with the proliferation of immature chondrocytes through SHH signaling. As expected, SAG21k treatment also increased the expression of the SHH targets *Gli1* and the receptor *Ptch1* (Figure 1). Local administration of SAG has been observed to increase chondrocyte proliferation in the cartilaginous fracture callus,⁴⁸ which is essential for callus bone formation. Thus, SAG21k promoted hedgehog signaling and chondrogenesis and would be expected to promote robust chondrogenesis to overcome impaired cartilage development in defect healing.

The second molecular pathway in the therapeutic approach, vitamin C, is an established regulator of bone and cartilage metabolism.^{44,49} PHD2 is the key PHD isoform for HIF-1 α regulation in many types of cells and knockdown of the *Phd2* gene results in embryonic lethality.⁵⁰ Chondrocyte-specific deletion of the *Phd2* gene results in an elevated number of Osx-expressing osteoblasts and a massive increase in trabecular BV

in the femoral secondary spongiosa.³³ *Phd2* is expressed in higher abundance in bone cells, where it mediates *Osx* expression.⁵¹ The inhibition of PHD2 would be expected to increase trabecular bone formation through angiogenic functions that mediate chondrocyte transdifferentiation.^{13,15}

We evaluated PHD2 regulation of chondrocyte differentiation in response to IOX2 in vitro (Figure 2). Expression of the markers of chondrocyte differentiation and angiogenesis was consistent with chondro-osteogenic regulation. *Phd2* expression was elevated, suggesting feedback regulation. EPO in particular was substantially increased by activated hypoxia signaling. High serum EPO levels are associated with adverse cardiovascular outcomes, but we have not measured the serum EPO in these mice and cannot eliminate this effect as a safety concern. These results suggest that PHD2 inhibition promotes osteoblast formation from chondrocytes through angiogenic functions.

A combined strategy that used SAG to promote cartilage development and IOX2 to subsequently promote the conversion of that cartilage to bone is based on evidence that chondrocytes stabilize endochondral healing bone injuries but also provide a source of osteoblasts for bone-related therapeutics.¹⁹ A temporal examination of *Col2*-expressing chondrocytes revealed a successive expression of markers of hypertrophic chondrocytes and osteoblasts, consistent with transdifferentiation of chondrocytes to osteoblasts.¹⁷ Lineage tracing has established that a significant proportion of growth plate and fracture callus osteoblasts are derived through transdifferentiation of terminally differentiated chondrocytes during development and repair,^{18,20} respectively, that proceeds at least partially through chondrocyte dedifferentiation and redifferentiation to osteoblasts.¹⁸ Disruption of *Phd2* expression in chondrocytes has produced a massive increase in trabecular bone mass by increased HIF-1 α -mediated chondrocyte hypertrophy and transdifferentiation into osteoblasts.³³ During endochondral bone repair, transdifferentiation at the cartilage–bone transition zone has been associated with pluripotent factor expression from the adjacent vasculature¹⁵ that would be expected from angiogenesis enhanced by PHD2 inhibition in the fracture tissues.

The unfractured contralateral femurs of each treatment group exhibited an apparent increase in the metaphyseal trabecular bone produced by the combined SAG21k:IOX2 therapy (Figure 3) that was evident in the parameters of trabecular bone formation (Figure 4). Because SHH has been observed to promote both chondrocyte^{17,23} and osteoblast⁵² proliferation and development, the increases in trabecular bone parameters in the absence of chondrocytes might have become more obvious through increased osteoblast functions mediated by SHH signaling and increased angiogenesis promoted by IOX2-mediated signaling. The combined systemic therapeutic approach would be expected to promote woven bone formation in a fracture callus, even without chondrocyte functions.

In segmental defect healing, the microCT results are consistent with the activation of bone formation expected from SHH and of angiogenesis from IOX2. Bony union was obvious in the group treated sequentially with SAG21k:IOX2 (Figure 3). There was a significantly greater BV in the SAG21k:IOX2-treated defects as compared to the other three groups (Figure 4). Interestingly, the IOX2 group displayed a reduced callus TV, which,

upon examination with microCT thresholds segmented to analyze the higher densities of cortical and remodeling bone, revealed a significant increase in the BV/TV bordering the open defect (Suppl Figure 4). This bone could have resulted from enhanced osteoblast commitment through IOX2 promotion of *Osx* expression and increased angiogenesis within the hard callus intramembranous bone.⁵³

An examination of the defect callus histology revealed that the bony union in the SAG21k:IOX2-treated group appeared superior to that of the individual therapies (Figure 5). Immunohistochemistry confirmed that the defect bone formation was associated with SAG21k and IOX2 functions. SAG21k applications increased the expression of SHH pathway genes *PTCH1* and *GLI1* in fracture cartilage at 5 days postsurgery, as well as the cartilage development transcription factor *SOX9* (Figure 6). By 4 weeks, IOX2 applications had increased *HIF-1α* and *VEGFA*, angiogenic factors that mediate bone formation (Figure 7). The individual SAG21k and IOX2 therapies did indeed enhance the expression of their respective pathway proteins in the target tissues at the anticipated postsurgery healing times and established that their sequential application might provide effective combined therapy.

This study has important limitations, most obviously the application of the small molecule activators in a single dosage, with each application restricted to a single and separate period of bone healing. It is probable that the dosages and times of application will require adjustment for maximum therapeutic benefit. We have also not yet demonstrated that the cartilage chondrocytes eventually transdifferentiate to osteoblasts and thereby promote osteogenesis. In the future, we will test more comprehensive dose-time applications of individual and combined therapies to study whether the activation of SHH signaling promotes skeletal stem cell proliferation and chondrocyte differentiation to increase cartilage development and whether hypoxia signaling promotes chondrocyte transdifferentiation to osteoblasts to increase new bone formation at the defect site. We will also extend the times allowed for bone formation up to 12 weeks postsurgery to allow complete examination of bone formation and remodeling, as each has been associated with SHH signaling in fracture repair.²⁸ This examination will also include mechanical testing of bone strength as a definitive measure of healing. Refinements for local applications of these small molecules might improve segmental defect healing but also minimize systemic side effects.

In summary, our studies have shown that a sequential systemic administration of SAG21k and IOX2 to promote hedgehog and hypoxia signaling, respectively, augmented bone formation within a segmental defect more than either therapy alone. We propose that SAG21k promoted SHH-regulated chondrocyte development to increase defect cartilage, which was then replaced by bone through IOX2 promotion of angiogenic functions. Because SAG21k and IOX2 are safe to use, sequential local applications present a safe and cost-effective therapeutic approach to promote healing in clinically challenging bone injuries.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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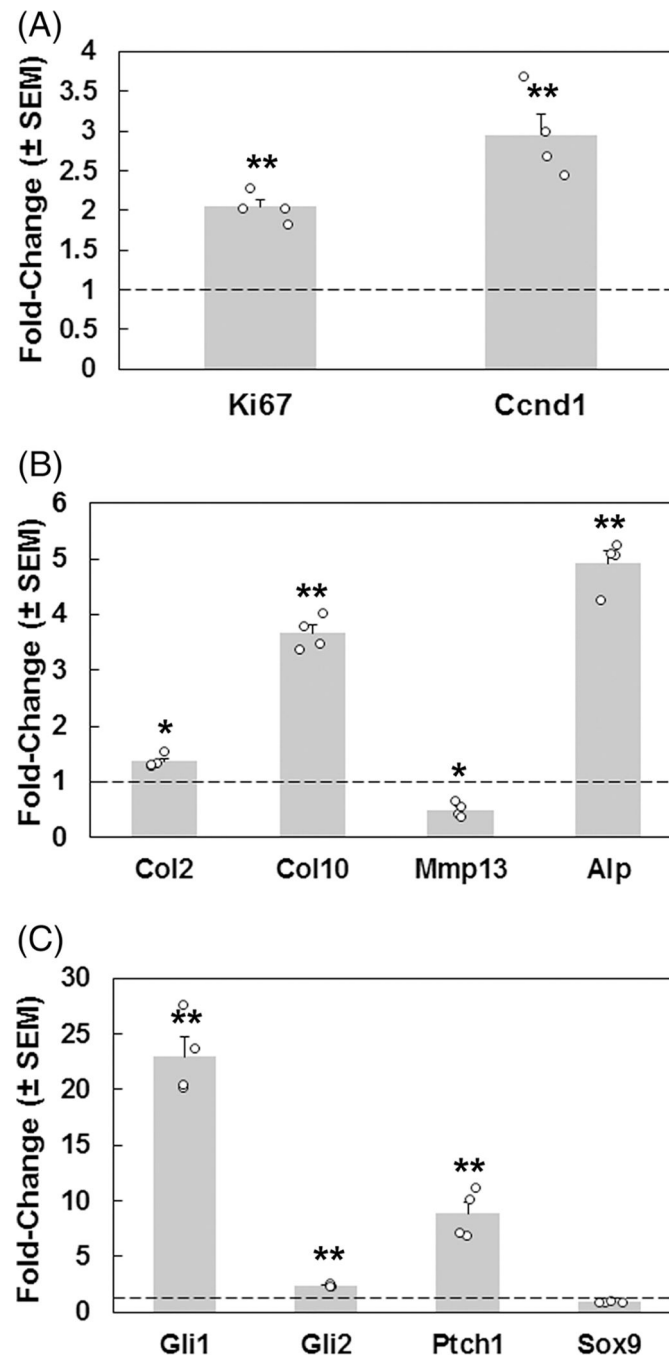
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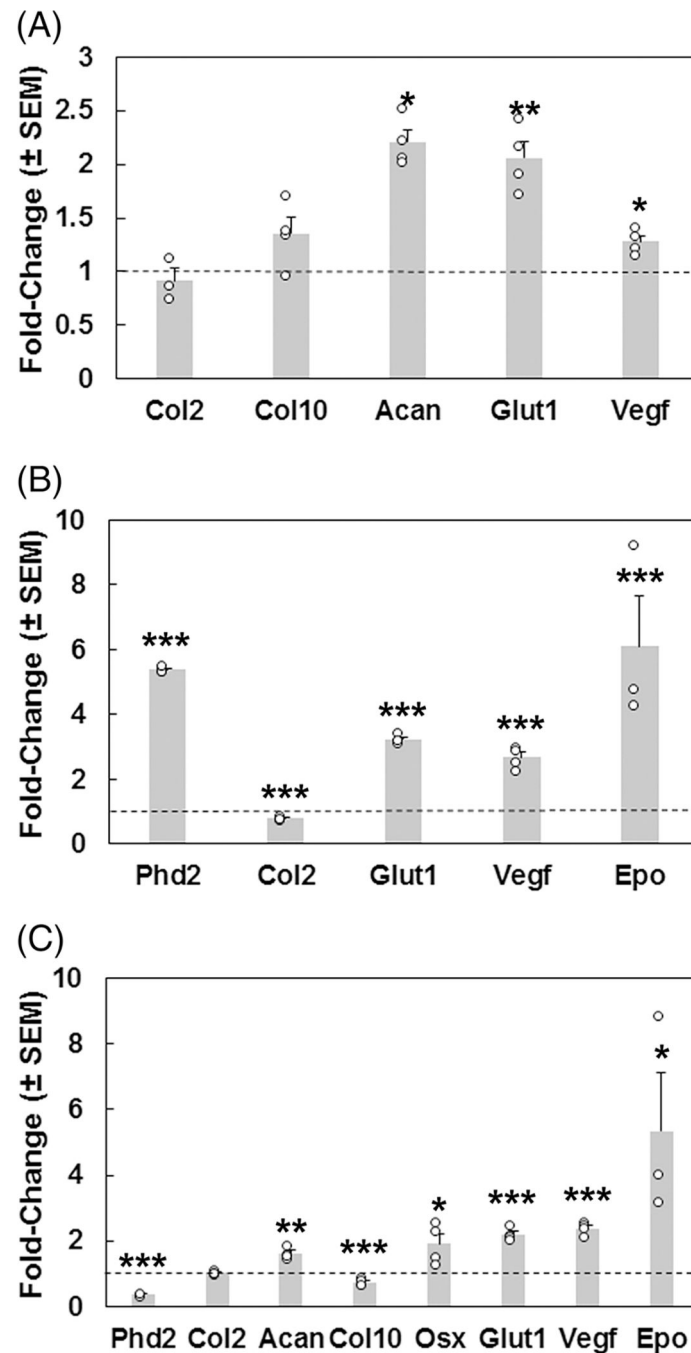
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**FIGURE 1.**

SAG21k induced the expression of proliferation-related genes (A), bone formation genes (B), and SHH pathway genes (C). SAG21k was incubated for 48 h with primary chondrocytes and gene expression was determined by real-time RT-PCR. (A) Proliferation genes were increased twofold to threefold in expression. The bone formation genes *Col10* and *Alp* were increased threefold and fivefold in expression, respectively. (B) The bone remodeling gene *Mmp13* was reduced in expression. (C) The expression of the SHH target genes *Gli1* and *Ptch1* was increased several fold. * $p < 0.01$; ** $p < 0.001$ versus vehicle.

RT-PCR, reverse transcription polymerase chain reaction; SAG21k, smoothened agonist 21k; SHH, sonic hedgehog pathway.

**FIGURE 2.**

IOX2 induced the expression of chondrocyte and cartilage matrix development, glycolysis, angiogenesis, and osteoblast genes. (A) In ATDC5 chondrocytes, *Col10* and *Acan* gene expression, which regulates chondrocyte and cartilage development, was significantly upregulated, as was the expression of the glycolysis gene *Glut1* and the angiogenic growth factor *Vegfa*. (B) In epiphyseal chondrocytes, the expression of the erythropoietic/angiogenic growth factor *Epo* was increased in response to IOX2, consistent with the inhibition of *Phd2* function. *Phd2* expression appeared upregulated. (C) Adenoviral-iCre deletion of *Phd2*

from chondrocytes isolated from *Phd2*-loxP mice confirmed that IOX2 inhibition of *Phd2* was similar to *Phd2* deletion in promoting the expression of these genes and the osteogenic transcription factor *Osx*. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$ versus adenoviral-GFP control. Acan, aggrecan; Col10, type-10 collagen; Epo, erythropoietin; GFP, green fluorescent protein; Glut1, glucose transporter 1; iCre, improved Cre; Phd2, prolyl hydroxylase domain; *Osx*, Osterix; Vegf, vascular endothelial growth factor.

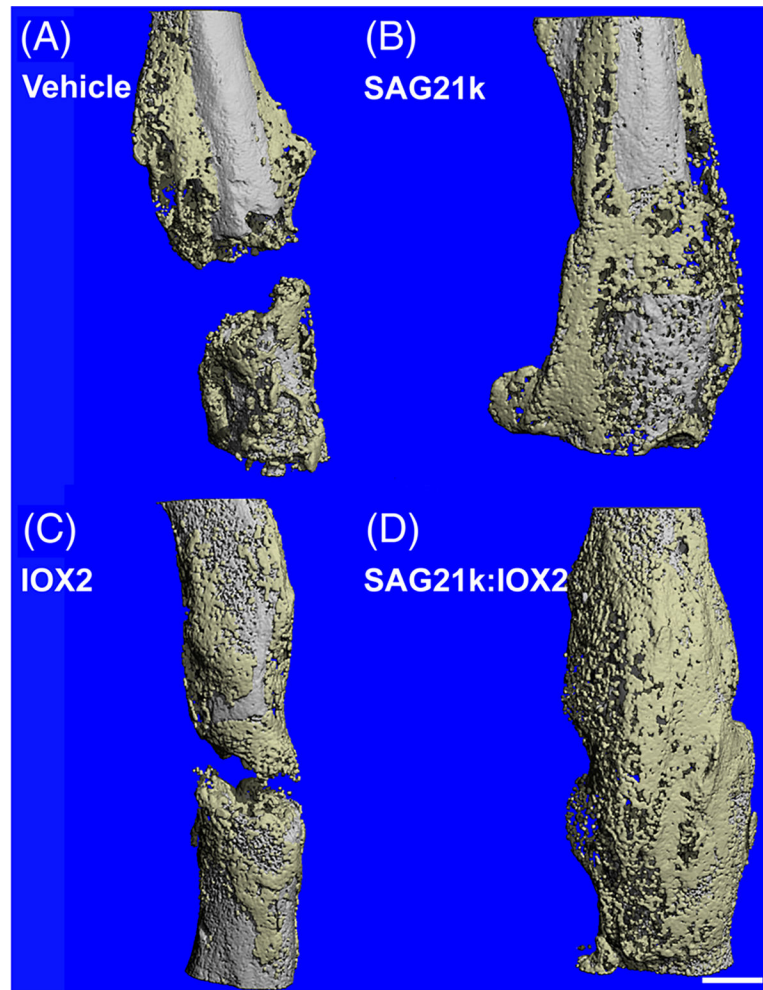
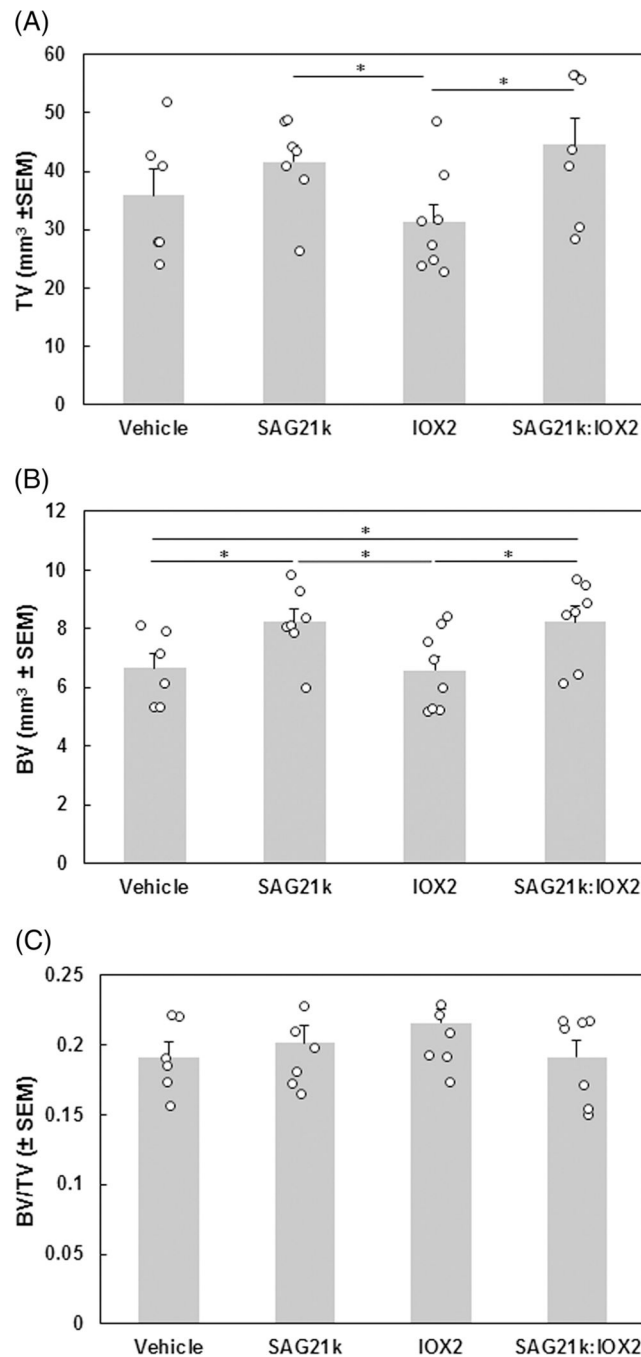


FIGURE 3.

Three-dimensional microCT reconstructions of segmental defects 6 weeks after surgery in mice receiving each of the therapies show that when compared to the vehicle control (A), the SAG21k therapy produced a larger callus that had not yet achieved bony union (B). (C) The IOX2 therapy produced a smaller defect callus without a bony union. (D) The SAG21k:IOX2 therapy produced a larger bony callus that had achieved union at the defect site. For this illustration, the entire callus was scanned and reconstructed. Scale bar = 1.0 mm. microCT, microcomputed tomography; SAG21k, smoothened agonist 21k. [Color figure can be viewed at wileyonlinelibrary.com]

**FIGURE 4.**

MicroCT analysis of defect callus total bone (250–1000 mg/cc) at 6 weeks postsurgery in mice receiving each of the therapies show that the SAG21k and combination SAG21k:IOX2 therapy increased the callus TV (A) and BV (B), with no difference in the BV/TV (C). Analysis was performed over a 2 mm scan centered at the defect midpoint. BV, bone volume; microCT, microcomputed tomography; SAG21k, smoothened agonist 21k; TV, tissue volume. * $p < 0.05$.

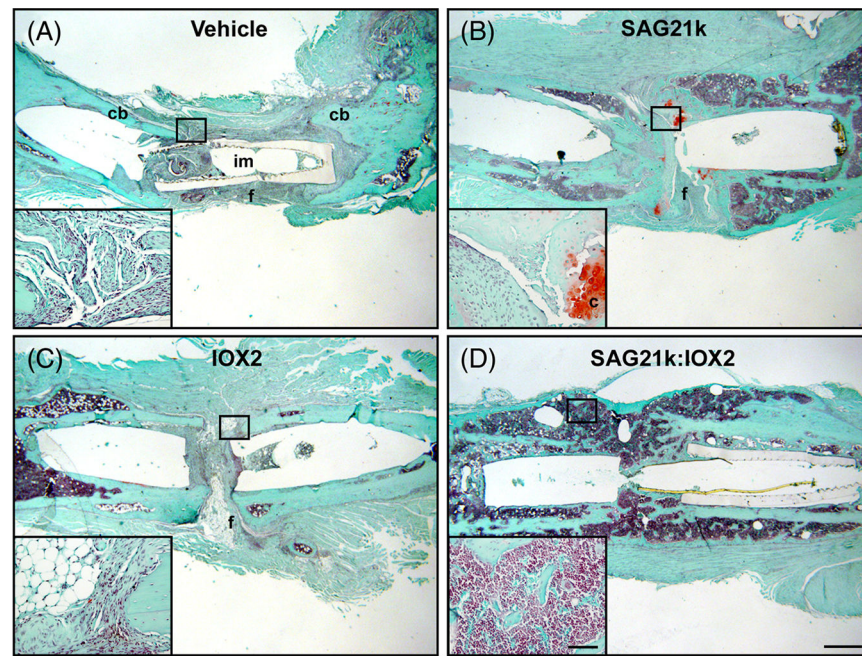


FIGURE 5.

Analysis of the segmental defect histology by Safranin-Orange staining at 6 weeks postsurgery. (A) Soft tissue remained in the defect in mice receiving the vehicle control. SAG21k therapy produced a larger callus with some cartilage and incomplete bony union (B), while IOX2 therapy resulted in a smaller bony callus without bony union (C). (D) SAG21k:IOX2 combination therapy promoted bony union at the defect. Scale bar = 1.0 mm. The box in the inset shows the location of the high magnification image; scale bar = 200 μ m. c, cartilage; cb, cortical bone; f, fibrous tissue; im, intramedullary space; SAG21k, smoothed agonist 21k. [Color figure can be viewed at wileyonlinelibrary.com]

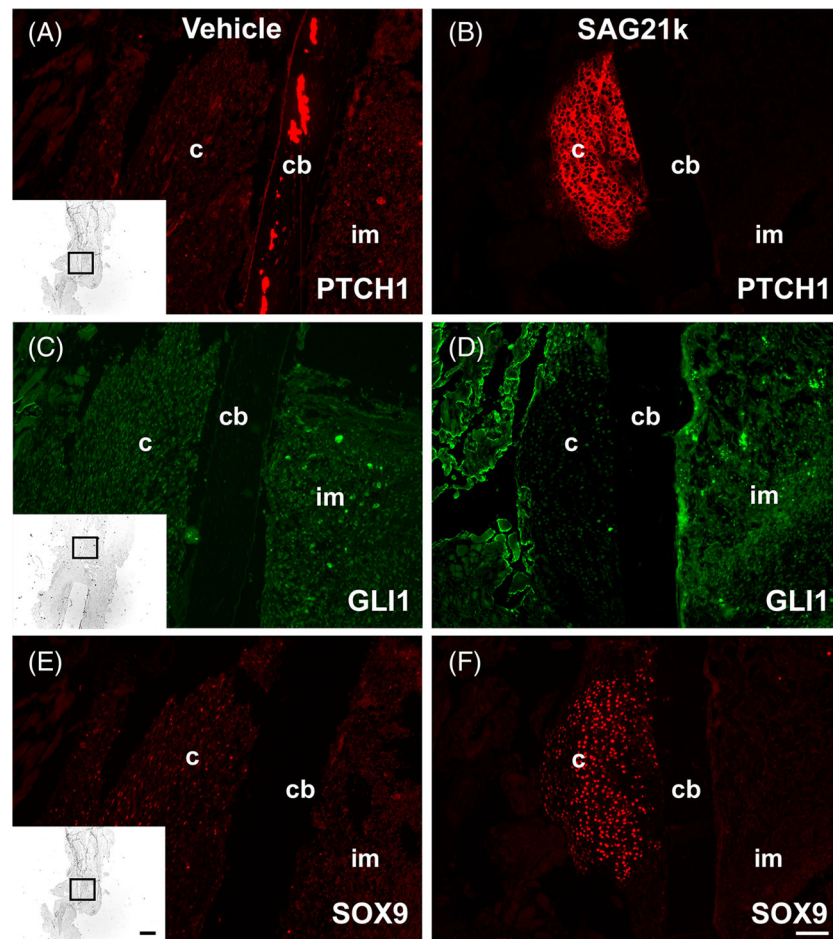


FIGURE 6.

SAG21k promoted an increased expression of SHH pathway targets in segmental defect cartilage at 5 days postsurgery. Vehicle control (A, C, E) was compared to SAG21k applications (B, D, F) at 6 weeks postsurgery for SHH pathway proteins PTCH1 (A, B), GLI1 (C, D), and the chondrocyte transcription factor SOX9 (E, F). Scale bar = 200 μ m. The box in the inset shows the callus location of the high magnification image; scale bar = 1 mm. c, cartilage; cb, cortical bone; im, intramedullary space; SAG21k, smoothed agonist 21k; SHH, sonic hedgehog pathway. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

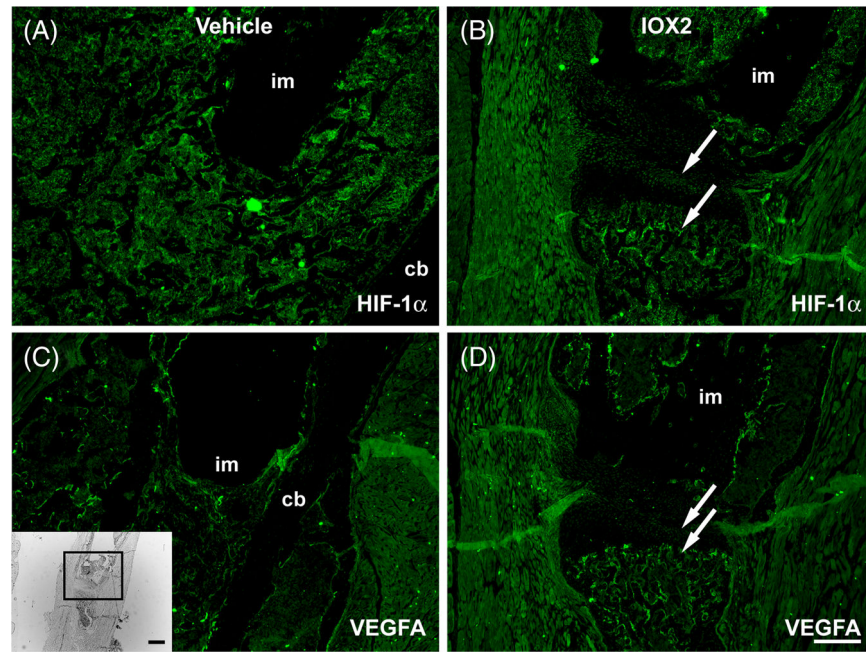


FIGURE 7.

IOX2 promoted an increased expression of angiogenic mediators in defects at 4 weeks postsurgery. Vehicle control (A, C) was compared to IOX2 applications (B, D) for HIF-1 α (A, B) and VEGFA (C, D) expression. Arrows indicate the expression of HIF-1 α (B) and VEGFA (D). Scale bar = 200 μ m. The box in the inset shows the callus location of the high magnification image; scale bar = 1 mm. cb, cortical bone; HIF-1 α , hypoxia-induced factor-1 α ; im, intramedullary space; VEGFA, vascular endothelial growth factor-A. [Color figure can be viewed at wileyonlinelibrary.com]